

# Shiva Panditaradhyula

Senior Python Engineer | Backend, Cloud & Platform Engineering

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## PROFESSIONAL SUMMARY

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Senior software engineer with 10+ years designing and delivering backend services, REST APIs, data pipelines, and cloud-native systems for enterprises including eBay, Progressive, Bank of America, and BNY Mellon. Currently build Python/Flask services for an enterprise storage-automation platform supporting Kubernetes and database workloads across data centers. Deep expertise in Python, Kubernetes, AWS and Azure, API security, data engineering, observability, and GenAI/RAG applications.

## TECHNICAL SKILLS

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**Languages:** Python, SQL, JavaScript, TypeScript, Bash/Shell, Scala, Perl, Java

**Backend & APIs:** Flask, FastAPI, Django/DRF, Celery, Node.js/Express, REST, GraphQL, WebSockets, SSE, OpenAPI/Swagger

**Cloud:** AWS (Lambda, API Gateway, EC2, S3, RDS, DynamoDB, Redshift, Glue, OpenSearch, Neptune, IAM), Azure, GCP

**DevOps & Containers:** Kubernetes, Docker, Helm, OpenShift, Terraform, Ansible, Jenkins, GitHub Actions, Azure DevOps

**Data Engineering:** Spark/PySpark, Databricks, Airflow, Hadoop, Hive, Pandas, NumPy, ETL pipelines

**Databases:** PostgreSQL, Oracle, MySQL, DB2, DynamoDB, MongoDB, Redis

**AI / GenAI:** LangChain, RAG, LLM agents, vector search (MongoDB Atlas, Azure AI Search), Microsoft Copilot Studio

**Security & Identity:** Okta, OAuth2, OIDC, SAML, JWT, Azure AD/Entra ID, RBAC, SCIM

**Observability:** Prometheus, Grafana, Splunk, OpenSearch, Azure Application Insights

**Storage & Systems:** Storage automation (NFS, Kubernetes persistent volumes), Linux (RHEL, Ubuntu, CentOS)

**Frontend & Testing:** React, TypeScript, Redux, Vue.js; Pytest, unittest, TDD

## PROFESSIONAL EXPERIENCE

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**eBay Inc., San Jose, CA**

**Oct 2025 – Present**

*Sr. Python Developer*

- Built a database-backed access-group management service in Python/Flask and PostgreSQL (8 REST endpoints for groups, membership, and access checks), replacing an external service dependency for authorizing self-service storage operations.
- Designed a manager-approval workflow for access-group membership changes, with approval and denial notifications to requesters and managers, supported across QA and production environments.
- Developed a REST API that reports the status of Kubernetes persistent volumes and their backing NFS storage for a partner platform team, with query filtering by volume and mount path.
- Fixed a cross-cluster matching defect in the volume-to-storage resolution logic that assigned the wrong storage cluster to about half of the results.
- Resolved production data-accuracy issues across regions and large clusters, including DNS-resolution fallbacks, data-freshness checks to prevent false decommissioned status, ownership disambiguation, and HTTP 413 response-size failures.
- Built a validation script that checks the API's cached results against live storage queries across all volumes, and shipped fixes through pre-production and production releases, backed by 3,000+ lines of Pytest unit and integration tests.

**Progressive**

**Aug 2023 – Sep 2025**

*Software Engineer / Sr. Python Developer*

- Built Python/Flask data-privacy services on OpenShift that scan enterprise databases for PII, including regex-based PII classification integrated into the enterprise metadata-scanning platform.
- Designed a Data Mart module that moves privacy scan results from Production, QA, and Dev into the enterprise data-privacy analytics platform.
- Built serverless REST and WebSocket APIs on AWS Lambda and API Gateway backed by DynamoDB (partition/sort-key design, TTL, encryption), and reduced Lambda cold-start latency.
- Secured APIs and React applications with Okta (OAuth2, OIDC, JWT validation, RBAC) and Azure AD/LDAP integration.
- Built LLM-powered features, including tool-using agents and Microsoft Copilot Studio copilots, with real-time streaming responses over Server-Sent Events.

- Built pipelines loading relational data into the AWS Neptune graph database and tuned AWS OpenSearch indexing for metadata search.
- Built accessible TypeScript/React interfaces (WCAG 2.1 AA) and Strapi headless-CMS APIs over REST and GraphQL; added automated tests to Jenkins and Azure DevOps CI/CD.

## **Hawaiian Airlines**

**Apr 2022 – Jun 2023**

### *Sr. Python Developer*

- Implemented Microsoft Entra ID single sign-on (MSAL) in React/TypeScript applications and automated user provisioning with Okta APIs and SCIM.
- Built real-time LLM response streaming in React over SSE and WebSockets with progressive rendering, error handling, and graceful fallback.
- Designed resilient REST APIs with OpenAPI specifications, retries with exponential backoff, circuit breakers, trace-ID propagation, and centralized error handling.
- Automated infrastructure with Terraform and Kubernetes deployments with Helm, autoscaling (HPA), rolling updates, and health checks.
- Built Pytest suites with reusable fixtures, parameterized tests, and markers for smoke and regression runs.

## **Comcast**

**Sep 2021 – Mar 2022**

### *Sr. Python Developer*

- Built FastAPI backend-for-frontend services that securely call Azure/Databricks-hosted ML models and expose governed APIs to React applications.
- Built MicroStrategy and Vue.js dashboards on SQL-backed APIs for leadership to track performance and operational metrics.
- Deployed multi-tier applications on AWS, configured IAM users and groups, automated S3 backups, and migrated on-premises databases to Redshift, RDS, and S3.

## **Prisma Health, Greenville, SC**

**Apr 2020 – Aug 2021**

### *Sr. Python Developer*

- Built Flask REST APIs with SQLAlchemy and PostgreSQL, using Celery with RabbitMQ and Redis for asynchronous tasks.
- Built an AWS data lake and CI/CD data pipelines with Glue, Lambda, and EC2, and scheduled workflows with Airflow.
- Sized, monitored, and tuned Databricks Spark clusters; wrote Sqoop change-data-capture jobs and Hive/Pig jobs for data cleaning.
- Built log-analysis, forecasting, and capacity-planning reports with Pandas, NumPy, scikit-learn, and Matplotlib.

## **Bank of America, New York, NY**

**Apr 2017 – Mar 2020**

### *Sr. Python Developer*

- Built Flask and Django REST microservices on AWS (EC2, EBS) with Docker, Jenkins, autoscaling, and Splunk monitoring.
- Built data pipelines with Flume, Pig, and Sqoop into HDFS, and Scala/PySpark jobs with Spark SQL that stored results in HBase.
- Automated provisioning and repetitive tasks with Terraform, Python, and Docker.
- Built an Airflow-based data validation framework; led root-cause analysis and production bug fixes.

## **BNY Mellon**

**Jan 2016 – Mar 2017**

### *Sr. Python Developer*

- Built internal web applications and microservices with Django, Flask, and MySQL.
- Automated server build and configuration with Ansible; set up AWS RDS with backups to S3.
- Built Python monitoring and notification tools and scripts that sped up access-request processing.

## **iHeartMedia**

**Jan 2015 – Dec 2015**

### *Software Developer*

- Built a Flask backend from scratch with RESTful APIs, and automated deployments with Fabric.
- Built a test harness tool with a Flask API and React front end; wrote SQL and PL/SQL for Oracle 12c and MS SQL Server.

## **EDUCATION**

**M.S., Computer Science** – Silicon Valley University

2015

**B.S., Mechanical Engineering** – Teegala Krishna Reddy Engineering College

2011